Inherited Social Trust and Depression Among Second-generation Immigrants*

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Abstract

This paper estimates the causal effect of culturally inherited social trust on the severity of depressive symptoms among second-generation immigrants. To isolate the impact of inherited social trust on depressive symptoms, I regress the depressive symptoms of second-generation immigrants against the average social trust in their parents' country of origin. Using a sample of second-generation immigrants residing in 22 European countries, I show that a one-standard-deviation increase in inherited social trust reduces the severity of depressive symptoms, as measured by the CES-D8 depression scale, by 6.5% of a standard deviation. Overall, this paper highlights a causal relationship between higher inherited social trust and less severe depressive symptoms among second-generation immigrants.

Keywords: Social capital, Trust, Depression, ESS, Second-generation immigrants
JEL Codes: I12, D13, D83, Z13

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1 Introduction

As reported by the World Health Organization (2017, 2022), mental health disorders have been rising worldwide over the last decade. In a recent global-scale study involving 29 countries and almost 160.000 participants, it emerges that about half of the participants face the risk of suffering from some form of mental illness during their lives (McGrath et al., 2023). According to the statistics from Eurostat (2021), the prevalence of chronic depression in Europe stood at approximately 7% in 2019. The Covid-19 pandemic exacerbated these figures, with depressive symptoms more than doubling in many European countries after 2020 (OECD and European Union, 2022). Extensive research in epidemiology, health policy, health economics, and other health-related disciplines has found a positive association between social capital and better health outcomes, including depression.[1] Yet, the effect of culturally transmitted social capital on the severity of depressive symptoms remains largely unexplored.

In this paper, I estimate the causal effect of inherited social capital on reducing the severity of depressive symptoms. Depressive symptoms are measured according to the Center for Epidemiologic Studies Depression Scale – CES-D8 (Radloff, 1977). Social capital is proxied by social trust, namely a frequently used proxy for cognitive social capital.[2] To isolate the effect of inherited social trust on the severity of depressive symptoms, I regress the depressive symptoms of second-generation immigrants living in Europe against the average social trust in the country of origin of their parents. This approach is referred to as the "epidemiological approach" (Fernández, 2011, p. 489) and rules out reverse causality as individual characteristics do not affect the average social trust in the parents' country of

[1] For recent meta-analyses and comprehensive literature reviews, for instance, see Ehsan et al. (2019), Rodgers et al. (2019) and Xue et al. (2020).

[2] As summarised by Cohen-Cline et al. (2018), the literature about social capital splits social capital into cognitive and structural. Indeed, according to Cohen-Cline et al. (2018, p. 162), cognitive social capital regards the individual perceptions about others' behaviours. On the other hand, structural social capital regards individuals' networking behaviours, such as being part of a community. In addition, some authors also consider a third dimension of social capital, namely relational social capital (e.g., see Claridge (2018)). In surveys, social trust is measured by questions such as *"Would you say that most people can be trusted or that you need to be very careful in dealing with people?"* in the Integrated Value Survey (IVS) (Haerpfer et al., eds, 2022; EVS, 2022).

origin. The results show that a one-standard-deviation increase in the average level of social trust in the country of origin of parents reduces the severity of depressive symptoms of second-generation immigrants by 6.5% of one standard deviation. The estimation is statistically significant at the 1% level and robust to different specifications. Moreover, I also show that social trust is more effective in reducing some of the symptoms included in the CES-D8 depression scale. Indeed, a one-standard-deviation increase in the social trust in the country of origin decreases the weekly depression by 8% of one standard deviation and weekly sadness by 6.8% of one standard deviation. On the other hand, inherited social trust does not affect other symptoms, such as feeling that everything is an effort and feeling restless.

The theoretical mechanisms through which higher social capital can enhance individual health are multifaceted. Epidemiologists argue that social capital can provide greater access to formal and informal healthcare, thereby positively affecting health outcomes (Kawachi and Berkman, 2000; Berkman and Glass, 2000). On the other hand, economists support the idea that social capital can improve individual health by reducing the adoption of risky behaviours (Folland, 2006, 2008). Both theories find empirical support: for instance, individuals with higher social capital have been shown to have greater access to healthcare services and to more information about healthcare programmes (Hollard and Sene, 2016; Debnath and Jain, 2020). Moreover, social capital has been shown to reduce the probability of risky health behaviours such as smoking cigarettes (Islam et al., 2017) and to promote "socially responsible behaviours" during the Covid-19 pandemic, which, in turn, led to fewer cases and deaths (Bartscher et al., 2021).

This paper contributes to the strand of literature estimating the causal effect of social capital on enhancing individual health.[3] For instance, using data from the Commonwealth of Independent States and an instrumental variable (IV) approach, d'Hombres et al. (2010) find a positive causal effect of social trust on self-reported health. Similar results have been

[3]It is important to highlight the existence of another strand of literature finding also a causal effect of health on social capital, thereby suggesting a simultaneous effect. See, for instance, Sirven and Debrand (2012), Rocco et al. (2014), Roychowdhury (2021), Lebenbaum et al. (2021) and Lebenbaum et al. (2024), among others.

found between informal social capital and self-reported health in Argentina (Ronconi et al., 2012). More recently, Cohen-Cline et al. (2018) show that social trust reduces the severity of depressive symptoms among twins in the U.S. However, the effect of inherited social trust on individual health and, more specifically, on the severity of depressive symptoms remains largely unexplored. In particular, applying the same methodology as this paper, Ljunge (2014a) is the only study showing that inherited social trust increases self-reported health among second-generation immigrants living in European countries. Nonetheless, self-reported health strongly reflects people's "vitality" (i.e., "feeling full of life and energetic"), rather than people's mental health (Au and Johnston, 2014, p. 27).[4] With this paper, I extend the previous literature on the effect of inherited social trust on health outcomes by shedding light on the role of inherited social trust in alleviating depression among second-generation immigrants living in European countries. To do so, I use the CES-D8 depressive scale as a measure of self-reported depression, which has been shown to be a valid measure of depressive symptoms for cross-country comparison (e.g., see Van de Velde et al. (2010) and Missinne et al. (2014)). Furthermore, I also separately analyse the effect of inherited social trust on each depressive symptom included in the CES-D8 depressive scale, showing that the symptoms that are mostly positively affected by inherited social trust are self-reported weekly depression and sadness.

The remainder of the paper follows the subsequent structure: Section 2 describes the data sources. Sections 3 and 4 explain the empirical strategy and present the econometric model. Section 5 presents the results. In Section 6 the robustness checks are shown. Section 7 concludes.

[4]Mental health is not even the second-best predictor of self-reported health. In fact, Au and Johnston (2014, p. 25) also shows that "physical functioning and bodily pain" predicts half of the effect of perceived vitality on self-reported health, whereas mental health predicts only one-quarter of the effect of perceived vitality on self-reported health.

2 Data

My analysis involves 5281 second-generation immigrants living in 22 European countries (including Russia and Israel) with parents originally from 58 different countries worldwide. I define second-generation immigrants as individuals born in the country of residence and whose parents were born in a different country. Due to sample size reasons, I follow the same procedure used by Daniele and Geys (2015) in assigning the country of origin of each second-generation immigrant. Precisely, I assign to second-generation immigrants their father's country of origin if their father or both their parents are immigrants. Otherwise, I assign to second-generation immigrants their mother's country of origin if only their mother is an immigrant.[5]

The final dataset combines individual- and aggregate-level data. The individual-level data provide information on respondents' mental health status, preferences, geographical location and socio-economic background (Subsection 2.1). The aggregate-level data provide information about economic and cultural variables in the country of origin of each respondent (Subsection 2.2). In Table A.1 of the Online Appendix A, I show that the final dataset does not differ significantly from the original dataset.[6] Hereafter, I describe the different data sources used in the main analysis and in the robustness checks.

2.1 Individual-level data

The individual-level data comes from wave 3 (European Social Survey (ESS 3), 2018), wave 6 (European Social Survey (ESS 6), 2023) and wave 7 (European Social Survey (ESS 7), 2018) of the European Social Survey (henceforth, ESS), as these waves include detailed questions about self-reported depressive symptoms. The ESS is a survey covering multiple European countries and has been released every two years since 2001. The survey aims to

[5]In the Online Appendix B, as a robustness check, I also replicate the main analysis by considering the mother's country of origin if the mother or both parents are immigrants and the father's country of origin if only the father is an immigrant. The results are similar to the ones in the main analysis.

[6]See also the Online Appendix A for more comprehensive descriptive statistics.

study the behaviours and attitudes of people living in Europe.[7] The waves 3, 6 and 7 of the ESS cover the years 2006-2007, 2012-2013 and 2014-2015, respectively. I include only countries having observations in at least two waves, such that the final sample of countries of residence consists of Belgium, Switzerland, Czech Republic, Germany, Denmark, Estonia, Spain, Finland, France, United Kingdom, Hungary, Ireland, Israel, Lithuania, Netherlands, Norway, Poland, Portugal, Russia, Sweden, Slovenia and Slovakia.

2.1.1 Depressive symptoms

The choice of waves 3, 6 and 7 of the ESS is motivated by the inclusion of eight survey questions about the respondents' experience of negative and positive emotions over the past week. Combined, these questions represent a "shortened version" of the Center for Epidemiologic Studies Depression Scale – CES-D8 (Radloff, 1977), which is used to capture respondents' severity of depressive symptoms over the past week.[8] The questions are: "*[...] please tell me how much of the time during the past week*": (i) "*you felt depressed?*"; (ii) "*you felt that everything you did was an effort?*"; (iii) "*your sleep was restless?*"; (iv) *you were happy?*"; (v) "*you felt lonely?*"; (vi) "*you enjoyed life?*"; (vii) "*you felt sad?*"; (viii) "*you could not get going?*". The answers all have the same structure, taking values going from 1 ("*None or almost none of the time*") to 4 ("*All or almost all of the time*"). I reverse the values of the scale for questions (iv) and (vi) so that higher values are associated with more severe depressive symptoms. From these questions, I create a depression scale defined as the sum of the scores from all the questions (i)-(viii), taking values from 8 (i.e., all answers are "*None or almost none of the time*") to 32 (i.e., all answers are "*All or almost all of the time*"). To facilitate the interpretation of the results in the empirical analysis, I standardise the scale with a mean equal to 0 and a standard deviation of 1. In addition, while the depression scale represents the main dependent variable, I will also consider each symptom in (i)-(viii) sepa-

[7] See the ESS webpage: `https://www.europeansocialsurvey.org/about-ess`.

[8] I borrow the term "shortened version" from the ESS website. For more info, see `https://www.europeansocialsurvey.org/findings/europeans-wellbeing/measuring-wellbeing`.

rately to better isolate the channels through which social trust affects depression. Similar to the depression scale, I standardise variables (i)-(viii) with a mean equal to 0 and a standard deviation of 1. Figure 1 shows the average non-standardised CES-D8 depression scale by country of residence.

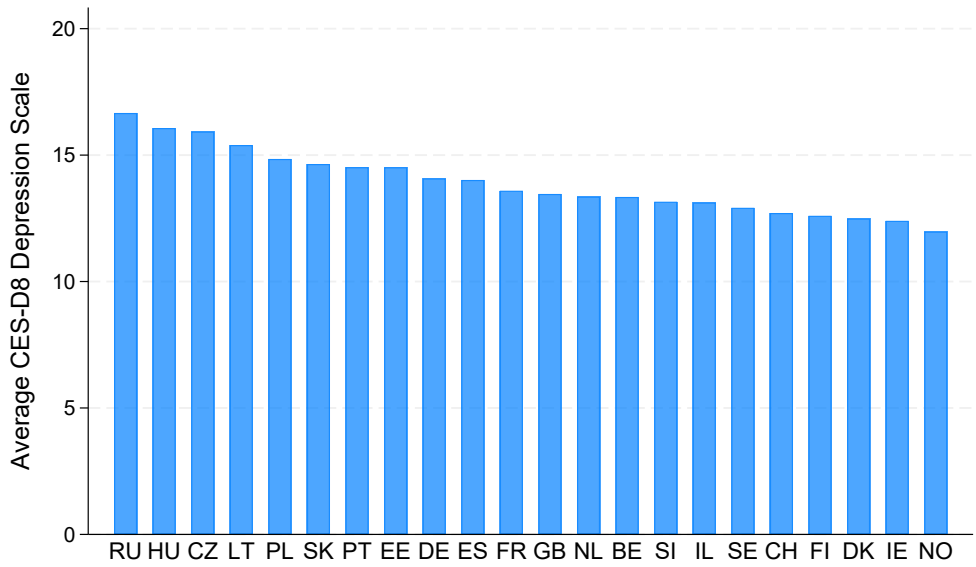

Figure 1: Average CES-D8 depression scale of second-generation immigrants by country of residence

Notes: the figure shows the average non-standardised CES-D8 depression scale for second-generation immigrants by country of residence.

2.2 Aggregate-level data

2.2.1 Social trust

The main independent variable is the level of social trust in the parents' country of origin. Social trust is computed using the Integrated Value Survey (IVS) (Haerpfer et al., eds, 2022; EVS, 2022).[9] The variable is the cross-country average of the answer to the survey question: "*Generally speaking, would you say that most people can be trusted or that you need to be*

[9]The Integrated Value Survey results from combining the World Value Survey (WVS) (Haerpfer et al., eds, 2022) and the European Social Survey (EVS) (EVS, 2022). See more information about the Integrated Value Survey (IVS) at `https://www.worldvaluessurvey.org/WVSEVStrend.jsp`.

very careful in dealing with people?". The variable takes the value 1 if the answer is "*Most people can be trusted*" and 0 if the answer is "*Need to be very careful*".[10] This variable has been used as a proxy for inherited social trust in previous research (e.g., see Algan and Cahuc (2010) and Figlio et al. (2019)). I average the social trust variable across all the available IVS waves. Furthermore, I standardise the scale with a mean equal to 0 and a standard deviation of 1 to facilitate the interpretation of the results in the empirical analysis.

2.2.2 Other aggregate controls

The other control variables about the country of origin of the parents come from multiple data sources, such as the IVS (Haerpfer et al., eds, 2022; EVS, 2022), the Hofstede's six dimension data matrix (Hofstede et al., 2005),[11] and the V-Dem Dataset (Coppedge et al., 2023), namely a collection of aggregate indices and country-level indicators from multiple data sources.[12] All the non-cultural variables are from the V-Dem dataset and have been averaged over the period 1950-1990, while the cultural variables from the IVS have been averaged using all the available waves.

The aggregate controls added to the model are a dummy variable for democracy from Cheibub et al. (2010),[13] a health inequality index from Pemstein et al. (2023), the infant mortality rate (per 1000 live births) from Gapminder (2018), the latent GDP per capita[14] from Fariss et al. (2022) to capture the level of economic development in the country of origin, individualism from the six dimensions cultural matrix by Hofstede et al. (2005), the average self-reported health, the average degree of religiosity and the average trust in the parliament as a proxy for average institutional trust from the IVS (Haerpfer et al., eds, 2022;

[10] Originally, in the IVS, the value of the answer "*Need to be very careful*" is 2. I recorded it at 0 to show the percentage of people who believe that most people can be trusted.

[11] For more info see `https://geerthofstede.com/research-and-vsm/dimension-data-matrix/`.

[12] All specific sources are cited below in correspondence with the specific variable, as suggested in the V-Dem guideline.

[13] The variable takes value 1 if a country has been a democracy at least half the time. Otherwise, the variable takes value 0.

[14] The use of latent GDP per capita from Fariss et al. (2022) is becoming more common because this dataset offers a larger coverage and relies on multiple data sources (e.g., see Leipziger (2023) and Leipziger (2024)).

EVS, 2022). These controls can potentially affect individual mental health and, following the recent literature applying the epidemiological approach (e.g., see Ljunge (2014a), Johansson et al. (2022) and Berggren et al. (2023)), are used only to test the robustness of the effect of inherited social trust on the severity of depressive symptoms in Subsection 5.1.

3 Methodology

In the next sections, I study the effect of inherited social trust on depression among second-generation immigrants. To do so, I regress the CES-D8 depression scale for the severity of depressive symptoms of second-generation immigrants on the social trust in the country of origin of their parents. In the economic literature, this approach is referred to as the "epidemiological approach" (Fernández, 2011, p. 489).[15]

As summarised by Fernández (2011, p. 490), the epidemiological approach relies on three key assumptions: first, culture is transmitted from parents to their offspring. Second, immigrants' beliefs and attitudes differ because of the transmitted ancestral cultural values. Third, immigrants residing in the same country are exposed to the same institutional and economic settings but to different culturally transmitted norms and values. Immigrant parents can shape the preferences and attitudes of their offspring by transmitting the cultural values acquired in their country of origin. This parents-to-offspring transmission of cultural values is referred to as "direct vertical socialisation" (e.g., see Bisin and Verdier (2001, p. 302) and Bisin and Verdier (2011, p. 341)). On the other hand, second-generation immigrants' traits and preferences do not impact the cultural values in the country of origin of their parents, thereby ruling out reverse causality. The DAG (Directed Acyclic Graph) in Figure 2 graphically summarises the conceptual framework of the epidemiological approach

[15]The epidemiological approach has been extensively used to study the causal effect of culture on different dependent variables in the fields of political economy and economic growth (e.g., see Algan and Cahuc (2010), Alesina and Giuliano (2010) and Daniele and Geys (2015)). More recently, this approach has been applied to a wider set of different research areas such as, for instance, education outcomes (Figlio et al., 2019; De Philippis and Rossi, 2021; Hanushek et al., 2022), economic and cultural integration (Berggren et al., 2023) and attitude towards climate change (Johansson et al., 2022).

when applied to inherited social trust and the severity of depressive symptoms.

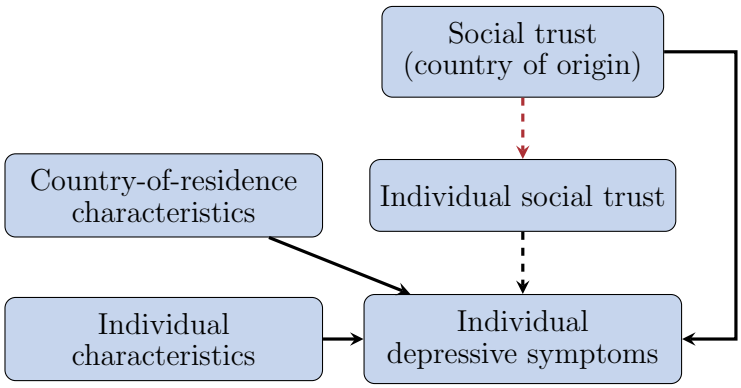

Figure 2: DAG for the epidemiological approach with inherited social trust

Notes: the figure shows the DAG for the epidemiological approach with inherited social trust. The graph shows that social trust in the country of origin of parents affects individual depressive symptoms of second-generation immigrants through direct vertical socialisation (red dashed arrow).

3.1 Descriptive evidence

The empirical evidence supports the hypothesis that social trust is transmitted to future generations. For instance, in the U.S., it has been shown that inherited social trust persists in immigrants for up to the fourth generation (Algan and Cahuc, 2010). Other evidence of intergenerational transmission of social trust has been documented in Germany (Dohmen et al., 2012) and among second-generation immigrants in European countries (Ljunge, 2014b). In the current analysis, the vertical transmission of inherited social trust among second-generation immigrants can be observed in the binned scatterplot in Figure 3. Figure 3 shows a positive correlation between the individual levels of trust of second-generation immigrants and the levels of trust in their parents' country of origin, thereby supporting the hypothesis that immigrant parents transmit the social trust acquired in their country of origin to their offspring.

As explained at the beginning of Section 3, the epidemiological approach also relies on the assumption that culturally transmitted values affect individual attitudes and preferences. Henceforth, I provide descriptive evidence that inherited social trust reduces the severity

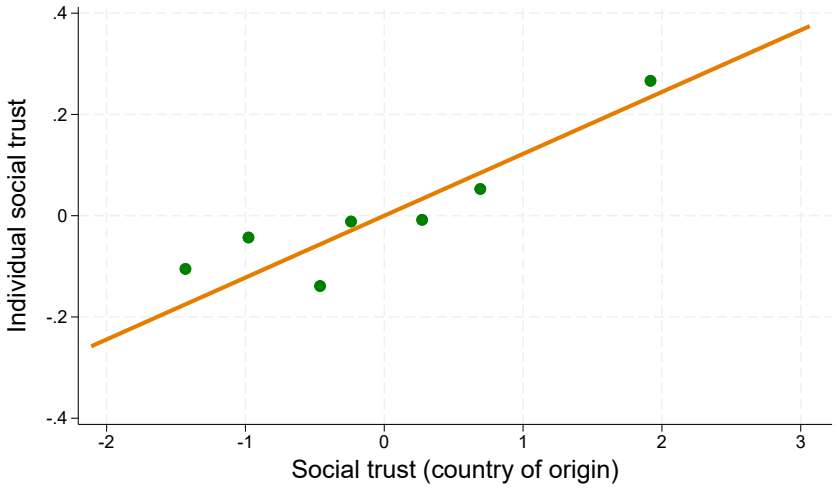

Figure 3: Binned scatterplot of individual social trust of second-generation immigrants regressed on social trust in the parents' country of origin.

Notes: the figure shows the binned scatterplot of individual social trust of second-generation immigrants regressed on social trust in the parents' country of origin against residence country-by-wave fixed effects. Individual social trust is measured on a scale going from 0 (You can't be too careful) to 10 (Most people can be trusted). Both social trust variables have been standardised with a mean of 0 and a standard deviation of 1. The coefficient of the regression is 0.059 (p-value<0.01). The adjusted-R^2 is 0.088. The number of bins in the scatterplot is computed using the integrated mean square error-optimal number of bins as suggested in Cattaneo et al. (2024). Binned scatterplot generated using the Stata command `binsreg` (Cattaneo et al., 2019)

of depressive symptoms among second-generation immigrants. Figure 4a shows the cross-country correlation between the average individual social trust and the severity of depressive symptoms of second-generation immigrants. The relationship is negative across countries, showing that countries with higher average individual social trust have, on average, less severe depressive symptoms. On the other hand, Figure 4b shows the cross-country correlation between the average severity of depressive symptoms among second-generation immigrants in their countries of residence and the average level of social trust in their countries of origin. Countries with higher average social trust in the country of origin of their second-generation immigrants show less severe depressive symptoms, supporting the hypothesis of the vertical transmission channel.

Figures 3 and 4 provide descriptive cross-country evidence showing that the data satisfies

the epidemiological approach assumptions. This evidence is purely descriptive and cannot be interpreted as causal. In the next section, I will apply the epidemiological approach to estimate the causal effect of inherited social trust on the severity of depressive symptoms.

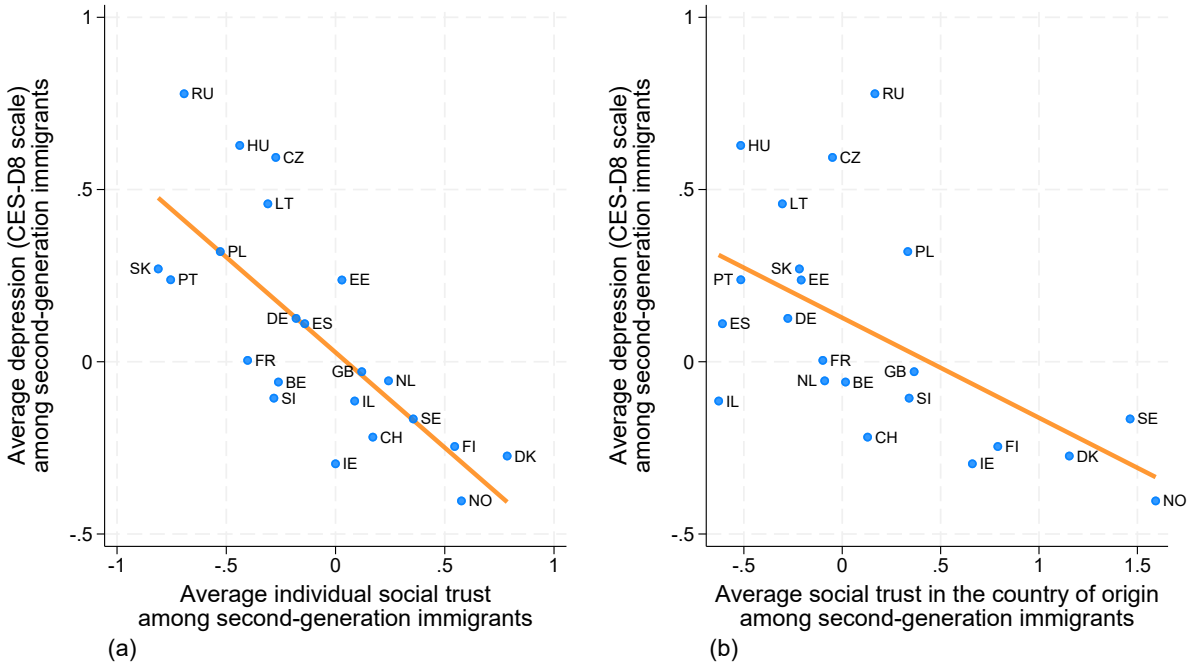

Figure 4: Cross-country relationship between social trust and severity of depressive symptoms

Notes: the plot shows the cross-country relationship between social trust and severity of depressive symptoms. Figure (a) shows that individual social trust is negatively correlated with the severity of depressive symptoms among second-generation immigrants. The raw correlation coefficient is -0.553 (p-value <0.01). Figure (b) shows that social trust in the country of origin is negatively correlated with the severity of depressive symptoms among second-generation immigrants. The raw correlation coefficient is -0.290 (p-value<0.01).

4 Empirical strategy

In the previous section, I provided descriptive evidence that, among second-generation immigrants, (i) parents transmit social trust to their offspring and (ii) countries with second-generation immigrants reporting less severe depressive symptoms are also characterised by

second-generation immigrants with higher individual social trust and higher levels of historical social trust in the country of origin. Henceforth, I apply the epidemiological approach to estimate the causal effect of inherited social trust on the severity of depressive symptoms among second-generation immigrants living in European countries. To do so, I estimate the following econometric model:

$$Depression_{ioct} = \beta_1 SocialTrust_{io} + \boldsymbol{X'}_{ioct}\boldsymbol{\gamma} + \theta_c \times \mu_t + \varepsilon_{ioct}, \tag{1}$$

where $Depression_{ioct}$ is the standardised measure of depression of an individual i with parents from the country of origin o and resident in the country c in wave t, where $c \neq o$. Depending on the specification, the variable can represent either the CES-D8 depression scale or a singular depressive symptom. On the right-hand side of (1), $SocialTrust_{io}$ is the standardised level of social trust for an individual i from parents' country of origin o, $\boldsymbol{X}_{ioct}$ is a set of individual controls including age, age squared, and a male dummy. These controls affects the severity of depressive symptoms but are unlikely to be affected by the social trust in the country of origin. $\theta_c \times \mu_t$ is a residence country-by-wave fixed effect to account for residence country-wave specific unobserved characteristics. Thus, I exploit the variation within each country of residence and not the variation across countries of residence to estimate the effect of inherited social trust on depressive symptoms. Standard errors are clustered at the country of origin level.

5 Main results

Table 1 presents the baseline model results showing how social trust in the country of origin affects the severity of depressive symptoms among second-generation immigrants. In all columns, I control for age, age squared and a male dummy. Column (1) shows the baseline model using the CES-D8 depression scale as the dependent variable. A one-standard-deviation increase in social trust in the country of origin decreases the severity of depressive

symptoms measured through the CES-D8 depression scale by 6.5% of a standard deviation. The effect is statistically significant at the 1% level.

The CES-D8 depression scale captures the severity of depression as it accounts for several symptoms related to depression; however, it is informative to understand which symptoms are more sensitive to inherited social trust. In Columns (2)–(9), I show the relationship between social trust in the country of origin and each depressive symptom included in the CES-D8 depression scale. First, the only two symptoms that are not statistically significantly correlated with social trust in the country of origin are *"Felt everything did as an effort"* and *"Sleep was restless"* in Columns (3) and (4). The other symptoms are all negatively correlated with social trust in the country of origin and statistically significant at the 1% level, with the exception of *"Felt lonely"*, which is statistically significant at the 5% level. Among the statistically significant symptoms, the ones with an effect that is larger than the CES-D8 depression scale effect are *"Felt depressed"* and *"Felt sad"*. Indeed, an increase in one standard deviation in social trust in the country of origin decreases these symptoms by 8% and 6.8% of a standard deviation, respectively. It is plausible that respondents were more likely to associate these two symptoms with depression.[16]

5.1 Robustness with individual and aggregate controls

In Table 2, I follow the recent epidemiological approach literature (e.g., see Ljunge (2014a), Johansson et al. (2022) and Berggren et al. (2023)) and I extend the analysis presented in Table 1 by including other individuals and country-of-origin aggregate controls to test the robustness of the results. The inclusion of individual and country-of-origin aggregate controls can raise endogeneity concerns; however, it is useful to test if social capital remains robust to different specifications.

In Table 2, the dependent variable is the CES-D8 depression scale, as it is the main dependent variable of the analysis. In Column (1), I show the baseline model from Table 1,

[16] In Table C.4 in the Online Appendix, I also use ordered probit and ordered logit for Columns (2)-(9) to test the robustness of the statistical significance of symptoms. Results do not change.

Table 1: Regression: severity of depressive symptoms and social trust in the country of origin (CO)

	CES-D8	**Depressive symptoms**							
	Depression scale	Felt depressed	Felt everything did as an effort	Sleep was restless	Felt lonely	Felt sad	Could not get going	Were unhappy	Did not enjoyed life
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Social trust (CO)	-0.065***	-0.080***	-0.019	-0.017	-0.036**	-0.068***	-0.045***	-0.055***	-0.032***
	(0.019)	(0.014)	(0.016)	(0.021)	(0.018)	(0.018)	(0.014)	(0.018)	(0.012)
Country-wave FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country of origin	58	58	58	58	58	58	58	58	58
Observations	5,281	5,281	5,281	5,281	5,281	5,281	5,281	5,281	5,281
Adjusted R-squared	0.059	0.053	0.034	0.025	0.039	0.064	0.060	0.040	0.047

Notes: Columns (1)–(9) report measures for the severity of depression regressed against social trust in the country of origin. The dependent variable is the CES-D8 depression scale in Column (1). Columns (2)-(9) present the effect of social trust in the country of origin on the different symptoms included in the CES-D8 depression scale. The name of the symptom is written on top of each column. In each regression, I control for age, age squared and a male dummy. The abbreviation "CO" stands for "Country of origin". Country-by-wave fixed effects are always included. Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

with no additional controls. From Columns (2) to (4), I progressively introduce individual and country-of-origin aggregate controls. The individual controls include occupation (1-digit ISCO-08), employment (paid work, unemployed, education, disabled, retired, other), a male dummy, the highest level of completed education (ISCED), age, age squared, type of respondent's domicile (big city, suburbs of a big city, town, country village or farm), marital status (married, divorced, widowed, none of these), degree of religiosity (measured on a 0-10 scale), a citizen dummy, a dummy taking value 1 if an individual i consider themselves as part of a discriminate group in the country c,[17] and a dummy taking value 1 if an individual was hampered by health problems in the last 12 months.[18] The aggregate controls from the country of origin are the same as listed in the Subsection 2.2.2, namely latent GDP per capita, health inequality index, infant mortality rate, a democracy dummy, individualism, religiosity, trust in the parliament, self-reported health.

In Column (2), I control for the individual controls. While individual controls may cause endogeneity problems, introducing these controls can help test the robustness of the relationship between social trust in the country of origin and the severity of depressive symptoms. The introduction of individual controls marginally reduces the size of the effect of social trust in the country of origin; yet, the coefficient remains statistically significant at the 1% level. Indeed, Column (2) shows that a one-standard-deviation increase in the social trust in the country of origin reduces the severity of depressive symptoms by 5.3% of a standard deviation.[19]

In Column (3), I include additional historical aggregate controls from the country of origin. Adding these controls can introduce endogeneity issues; however, introducing other historical

[17] The question in the ESS is: "*Would you describe yourself as being a member of a group that is discriminated against in this country?*".

[18] Value 1 is assigned if a respondent must report they had at least one of these health problems: heart or circulation problems, high blood pressure, breathing problems, allergies, back or neck pain, muscular or joint pain in hand or arm, muscular or joint pain in foot or leg, stomach or digestion related problems, skin condition related problems, severe headaches and diabetes.

[19] Individual income is not included because waves 1–3 of the ESS do not have an income scale that is comparable with the subsequent waves. Nevertheless, (i) in the model, some variables are strongly correlated with income, such as education, occupation and employment and (ii) in the Online Appendix C (Table C.1) I also consider only waves 6 and 7 by adding the income variable and I show that results are robust.

aggregate variables in the baseline econometric model serves as an informative robustness check to verify whether social trust in the country of origin does not capture the effect of other aggregate variables on the severity of depressive symptoms. The inclusion of the aggregate controls increases the effect of social trust in reducing the severity of depressive symptoms among second-generation immigrants. Indeed, a one-standard-deviation increase in the social trust in the country of origin reduces the severity of individual depressive symptoms by 9.2% of a standard deviation. The coefficient is statistically significant at the 1% level.[20]

Finally, in Column (4), I add both individual and aggregate controls. A one-standard-deviation increase in the social trust in the country of origin results in a reduction of the severity of individual depressive symptoms by 8% of a standard deviation. The coefficient is statistically significant at the 1% level. In addition, in Table C.2 in the Online Appendix, I replicate the same exercise for the other symptoms that are statistically significant at least at the 5% level in Table 1. Only *"Were unhappy"* and *"Did not enjoy life"* were not robust to the introduction of other aggregate controls from the country of origin.

6 Other robustness checks

In this section, I provide other tests to assess the robustness of the main results shown in Table 1. Precisely, I consider four major problems that can undermine the causal estimation of the relationship between CES-D8 depression scale and social trust in the country of origin in Table 1: (i) geographical and cultural self-selection bias; (ii) country-specific trends; (iii) cluster heterogeneity; (iv) volatility of the variable social trust in the IVS dataset.

[20] In Table C.3 in the Online Appendix C, I also show how the estimates change by introducing only one aggregate control at a time. Social trust in the country of origin is statistically significant at the 1% level for all the variables except for latent GDP per capita, individualism and infant mortality rate, which are statistically significant at the 5% level. Only political trust is statistically significant at the 1% level. The other aggregate variables are always statistically insignificant.

Table 2: Regression: CES-D8 depression scale and social trust in the country of origin (CO) with controls

	CES-D8 depression scale			
Variables	(1)	(2)	(3)	(4)
Social trust (CO)	-0.065***	-0.053***	-0.092***	-0.080***
	(0.019)	(0.016)	(0.031)	(0.028)
Country-wave FE	✓	✓	✓	✓
Individual Controls		✓		✓
Country-of-origin controls			✓	✓
Country of origin	58	58	58	58
Observations	5,281	5,281	5,281	5,281
Adjusted R-squared	0.059	0.152	0.058	0.152

Notes: Columns (1)–(4) report the depression symptoms regressed on social trust in the country of origin (CO), age, age squared and a male dummy (1), other individual controls (occupation, employment, the highest level of completed education (ISCED), type of respondent's domicile, marital status, degree of religiosity, a citizen dummy, a discrimination dummy, and a hampered-by-health-problems dummy) (2) and country-of-origin controls (latent GDP per capita, health inequality index, infant mortality rate, democracy dummy, individualism, religiosity, trust in the parliament, self-reported health) and both individual and aggregate controls (4). Country-by-wave fixed effects are always included. The abbreviation "CO" stands for "Country of origin". Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

6.1 Geographical and cultural self-selection

The literature on migration theorises and empirically shows that immigrants self-select themselves into the destination countries (e.g., see Borjas (1987) and Borjas et al. (1992, 2019)). Hence, I follow Hanushek et al. (2022) and, in Table 3, I show four robustness checks testing whether the relationship between social trust in the country of origin and the CES-D8 depression scale presented in Table 1 is affected by migration selectivity. In Column (1), I control for the continent-of-origin fixed effect to account for cultural heterogeneity and migration selectivity within and between continents. The five continents are Africa, America, Asia, Europe and Oceania, as defined in the CEPII's GeoDist database (Mayer and Zignago, 2011). By including the continent-of-origin fixed effect, I do not find a significant variation in the size of the coefficient of social trust in the country of origin, which remains statistically

significant at the 1% level.

In Column (2), I control for "*cultural variance*" (Hanushek et al., 2022, p. 2303). If immigrant selectivity is correlated with an average cultural value in the country of origin, such correlation could represent a source of bias in the analysis. According to Hanushek et al. (2022), this bias should be more accentuated in countries of origin with a large cultural variance. In Column (2), I include the standard deviation of social trust in the country of origin as a measure of cultural variance. The measure for cultural variance is not statistically significant, and the size of the country-of-origin social trust coefficient does not change significantly with respect to the main model in Table 3. On the other hand, social trust in the country of origin becomes statistically significant at the 5% level. This result could also be driven by the within-waves volatility of the variable social trust in the IVS dataset (Zanella and Bellani, 2024); however, in Subsection 6.4, I show that social trust volatility does not represent a major concern in this analysis, thereby reducing the possibility that the volatility problem drives results in Column (2).

In Columns (3) and (4), I control for the distance between the country of residence and the country of origin to account for distance-related immigrant selectivity. I control for two types of distances: geographical distance and cultural distance. The geographical distance is the spatial distance between the capital in the country of origin and the capital in the country of residence. Data about the latitude and longitude comes from the CEPII's GeoDist database (Mayer and Zignago, 2011). The cultural distance is the absolute distance between the level of social trust in the country of origin and the country of residence. Neither the cultural nor the geographical distance are statistically significant or significantly affect the estimation of social trust in the country of origin. As a result, the effect of social trust in the country of origin on the CES-D8 depression scale in Table 1 is robust to geographical and cultural self-selection bias.

Table 3: Regression: CES-D8 depressive symptoms, geographical and cultural self-selection

	CES-D8 depression scale			
			Migration distance controls	
Variables	Continent-of-origin fixed-effect (1)	Cultural variance (2)	Geographical distance (3)	Cultural distance (4)
Social trust (CO)	-0.068***	-0.066**	-0.063***	-0.071***
	(0.019)	(0.031)	(0.019)	(0.019)
Social trust SD (CO)		0.030		
		(0.619)		
Geographical distance			0.000	
			(0.000)	
Social trust (CO) × Cultural distance				0.003
				(0.004)
Country-wave FE	✓	✓	✓	✓
Country of origin	58	58	58	58
Observations	5,281	5,281	5,281	5,281
Adjusted R-squared	0.059	0.058	0.058	0.058

Notes: Columns (1)–(4) report the depression symptoms regressed on social trust in the country of origin (CO), while controlling for continent-fixed effect (1), the standard deviation of social trust in the country of origin (2), geographical distance (3) and cultural distance (4). Age, age squared, male dummy, and country-wave fixed effects are always included. All aggregate variables are demeaned. The abbreviation "CO" stands for "Country of origin". Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

6.2 Country-specific trends

Another problem is the presence of country-specific trends driving the results. For instance, some countries of residence might be outliers in terms of average depression, while some countries of origin might be outliers in terms of historical social trust, thereby driving the results. To rule out this concern, I re-estimate the effect of social trust on the severity of depressive symptoms by performing a leave-one-country-out robustness test. I repeat this exercise twice: once for countries of origin and once for countries of residence. Figure 5 shows the estimated coefficients for social trust in the country of origin, as in the empirical specification of Column (1) of Table 1. Each coefficient is estimated by leaving only one country out of the sample at a time. When a country is excluded from one estimation, the

country is re-included again in the other estimations. If the estimated coefficients present large differences in their size or statistical significance, then country-specific trends or outliers driving the results might exist. As shown in Figure 5, social trust is always statistically significant at the 5% level for countries of origin and residence. Additionally, the sizes of the coefficients do not vary significantly. The only model that shows a lower coefficient — although still statistically significant with 95% confidence intervals— is represented by the one excluding the Netherlands as country of origin. However, the Netherlands is not an outlier in terms of the average historical social trust (see Figure A.3 in the Online Appendix). As a consequence, the analysis in Figure 5 shows that the findings in Table 1 remain robust against country-specific trends or outliers.

6.3 Cluster heterogeneity

A second concern pertains to the presence of heterogeneous clusters. Studies applying the epidemiological approach often feature countries of origin with a large number of observations, while others with only a few observations, or even just one.[21] This analysis is no exception, as countries of origin such as Russia, Italy and Germany represent 17.23%, 9.43% and 8.65% of the total sample, while other countries are represented by only one or a few observations (See Table A.3 in the Online Appendix). This unequal distribution of observations across clusters can have large effects on the standard errors. Indeed, cluster heterogeneity may cause two problems: the first pertains to the leverage and influence that larger clusters can have on the final results (MacKinnon et al., 2023b). In simple terms, high-leverage clusters are clusters with a high volume of information that can significantly affect the OLS estimation.[22] Removing one of these clusters from an OLS model could potentially change the OLS estimation. The second problem pertains to the fact that an OLS model without cluster fixed effect is more likely to suffer from intra-cluster dependence, which has a large

[21] See, for instance, the appendix of Alesina and Giuliano (2011), Ljunge (2014b), Figlio et al. (2019), De Philippis and Rossi (2021) and Berggren et al. (2023).

[22] Figure D.1 in the Online Appendix shows the unequal leverage and partial leverage of each cluster.

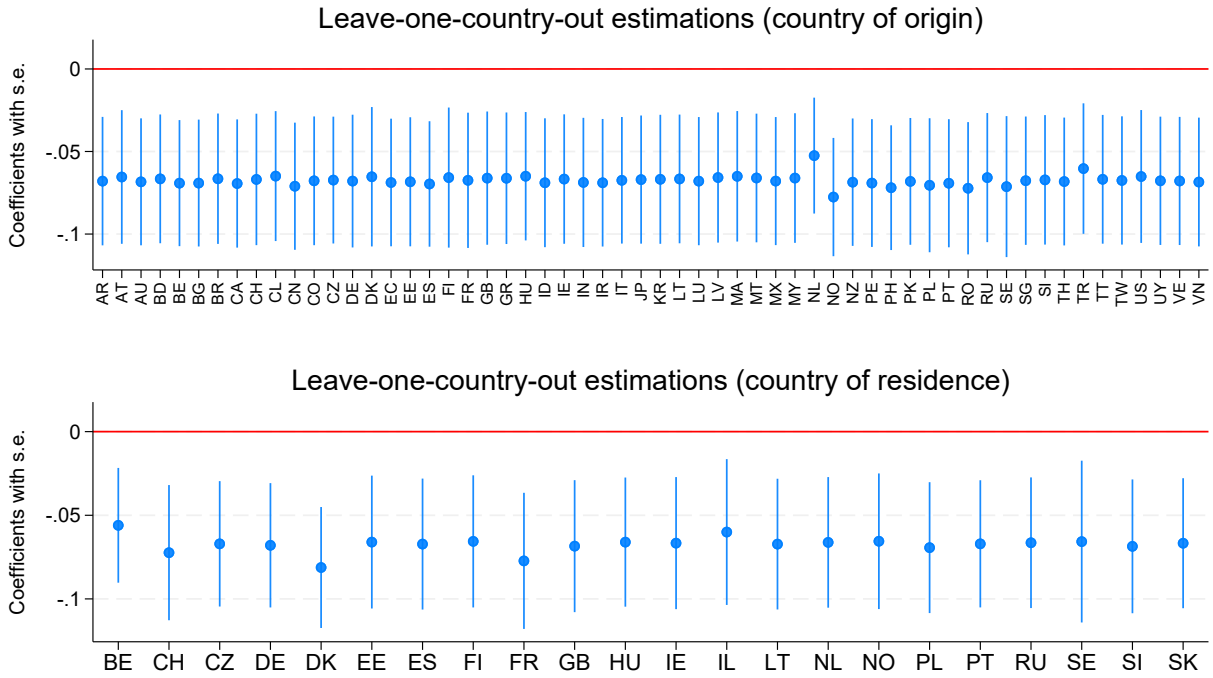

Figure 5: Estimation of the regression coefficients leaving one country out

Notes: the figure shows the estimation of coefficients with 95% confidence intervals obtained by regressing the CES-D8 depression scale against social trust in the country of origin. The first graph (above) shows the estimated coefficients for social trust by leaving out one country of origin at a time; the second graph (below) shows the estimated coefficients for social trust by leaving out one country of residence at a time. In each estimation, the ISO-2 code of the omitted country appears on the x-axis. Age, age squared, male dummy, and country-wave fixed effects are always included. Standard errors are clustered at the country-of-origin level.

effect on standard errors when some clusters are large (MacKinnon et al., 2023a).

To test the robustness of results to the above-mentioned problems, I follow the guide provided by MacKinnon et al. (2023a) which suggests comparing the CV_1 cluster-robust variance estimator (CRVE) — which corresponds to the one estimated in the standard specification in Table 1—, the CV_3 CRVE — that is a jacknife estimator— and the restricted wild-cluster (WCR) bootstrap estimator.[23] In addition, I also apply the randomization-t (RI-t) and the

[23]For the CV_1 and CV_3 I use the `bootcluster` command in Stata (MacKinnon et al., 2023b). For the WCR bootstrap estimator, I use the `boottest` command in Stata (Roodman et al., 2019). The `boottest` command does not report the standard errors, but only the p-value and the confidence intervals. The number of interactions equals 999999.

randomization-c (RI-c) techniques (Young, 2019).[24] Results are shown in Table 4 for the CES-D8 depression scale (i.e., Column (1) of Table 1). All the re-estimated p-values are of similar size and consistently smaller than the 1% statistical level. As a consequence, it is plausible to infer that the relationship between the CES-D8 depression scale and social trust in the country of origin in Table 1 is robust to cluster heterogeneity.

Table 4: Robustness: heterogeneous clusters with CES-D8 depression scale as dependent variable

Panel (a)	**Estimation:** $\widehat{\beta}_1$			
	t-statistics	Stand. Err.	P-value	Confidence Interval
Clustering levels: method	(1)	(2)	(3)	(4)
Country of origin: CV_1	-3.4526	0.0190	0.0011	[-0.1021 , -0.0271]
Country of origin: CV_3	-2.9705	0.0224	0.0043	[-0.1082 , -0.0211]
Country of origin: WCR bootstrap	-3.4526	—	0.0038	[-0.1074 , -0.0218]
Panel (b)	**Estimation:** $\widehat{\beta}_1$			
	Baseline p-value	Minimum p-value	Maximum p-value	Randomised p-value
Clustering levels: method	(1)	(2)	(3)	(4)
Country of origin: RI-t	0.0011	0.0030	0.0031	0.0030
Country of origin: RI-c	0.0011	0.0000	0.0000	0.0000

Notes: the table shows the p-values of the estimation of social trust in the country of origin in Column (1) of Table 1 estimated using different bootstrapping and randomisation techniques. In Panel (a), the CV_1, CV_3 and WCR bootstrap estimators (MacKinnon et al., 2023a) are presented. The number of interactions equals 999999. In Panel (b), the randomisation-t (RI-t) and randomisation-c (RI-c) estimators (Young, 2019) are presented. The number of interactions equals 10000.

[24] I use the `randcmd` command in Stata (Young, 2020). The `randcmd` command for computing randomization inference p-values provides two types of randomization: randomization-t (RI-t) and randomization-c (RI-c). Randomization-t (RI-t) "corresponds to bootstrap tests based on the percentiles of Wald statistics" (Young, 2019, p. 574), while randomization-c (RI-c) "corresponds to bootstrap tests which use the distribution of bootstrapped coefficients to calculate the covariance matrix" (Young, 2019, p. 574). As the `randcmd` is computationally less efficient than `boottest`, I used a smaller number of repetitions than with the `boottest` command, that is 10000. This number of interactions is the same as in Young (2019).

6.4 Social trust measure volatility

A final concern regards deriving proxies of historical cultural values from survey questions. Proxying time-persistent cultural values using the IVS dataset is common in the literature applying the epidemiological approach (e.g. see Alesina and Giuliano (2010) and Fernández (2011), among others). However, more recently, Zanella and Bellani (2024) and Zanella (2024) show that using survey questions to capture the time-persistence of cultural values may be subject to within- and across-waves volatility. The authors show that this volatility of survey variables can reduce the robustness and reliability of the proxy. Although Zanella and Bellani (2024) also show that the variable social trust from the IVS is not heavily affected by the wave volatility, in this Subsection, I provide evidence that the volatility problem does not substantially alter the main results of the paper.[25]

In Table D.1 in the Online Appendix, I replicate the results of Table 1 by averaging social trust first over waves 1-3 (1981-1998) —which present lower volatility (Zanella and Bellani, 2024)— and then over waves 4-7 (1999-2022). Results using waves 4-7 are similar to the main model in Table 1, while results using waves 1-3 present small differences. Indeed, the estimation for the CES-D8 depression scale is about 20% smaller and statistically significant at the 5% level. While looking at the symptoms, no difference in statistical significance is found for the main drivers (i.e., *"Felt sad"* and *"Felt depressed"*), even though they are about 20% smaller in size. The statistical significance of less robust symptoms, such as *"Were unhappy"*, *"Could not get going"*, and *"Did not enjoy life"* reduces to 5%, while the symptom *"Felt lonely"* becomes statistically insignificant.

Although some of the results estimated using waves 1-3 are smaller in size or statistically weaker, it is important to highlight that the estimations for waves 1-3 might differ due to sample variations, as some countries appear in the IVS only starting from wave 4. Indeed, the sample size is reduced by about 10% (from 5,281 to 4,737) and the number of countries of origin by about 20% (from 58 to 47). In Figure D.2, I also show that for all the variables,

[25]Social trust is probably more stable because it is captured by a binary scale rather than an ordinary scale. Indeed, ordinary scales are more sensitive to time-volatility (Zanella and Bellani, 2024).

the coefficient estimations are not different among the different models, as the confidence intervals always overlap. Moreover, the results that are mostly affected by survey volatility are those symptoms with a smaller or less robust effect in Table 1. To further show that the results in Table D.1 are affected by sample size rather than survey volatility, in Table D.2 I replicate the analysis using social trust averaged over waves 1-7, but with the reduced sample size of the model that uses social trust averaged over waves 1-3. The results are similar to the ones obtained using only waves 1-3, thereby suggesting that the smaller sample size drives the weaker statistical significance and the smaller size of the estimates and survey volatility does not represent a major concern.

7 Conclusion

This paper estimates the causal effect of inherited social trust in reducing the severity of depressive symptoms among second-generation immigrants living in Europe. Using a sample of second-generation immigrants residing in 22 European countries, I show that a one-standard-deviation increase in inherited social trust reduces the severity of depressive symptoms, as measured by the CES-D8 depression scale Radloff (1977), by 6.5% of a standard deviation. Among the symptoms included in the CES-D8 depression scale, inherited social trust has a larger effect on questions of the CES-D8 depression scale capturing weekly depression and weekly sadness. Indeed, a one-standard-deviation increase in the social trust in the country of origin decreases weekly depression by 8% of one standard deviation and weekly sadness by 6.8% of one standard deviation. Conversely, inherited social trust does not affect other symptoms, such as feeling that everything is an effort and feeling restless in the last week. I interpret these results as a possible consequence of the fact that people are more likely to associate questions about weekly sadness and weekly depression with depressive symptoms.

This paper contributes to the empirical literature estimating the causal effect of social capital on health outcomes. Specifically, this paper expands previous research on the effect

of inherited social trust on health (e.g., see Ljunge (2014a)) by focusing on the severity of depressive symptoms rather than self-reported or subjective health. The focus on inherited social trust in mental health is crucial to understanding the cultural determinants of the recent rise in mental health disorders. Moreover, a better understanding of how culturally transmitted values affect health outcomes can provide empirical guidance for policymakers to implement more effective policy interventions. For these reasons, future research should continue investigating the role of culturally transmitted social capital on different health dimensions, such as physical health (e.g. chronic physical health conditions) or other specific mental health variables (e.g. anxiety).

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Inherited Social Trust and Depression Among Second-generation Immigrants

(ONLINE APPENDIX — Supplementary material.)

A Descriptive statistics

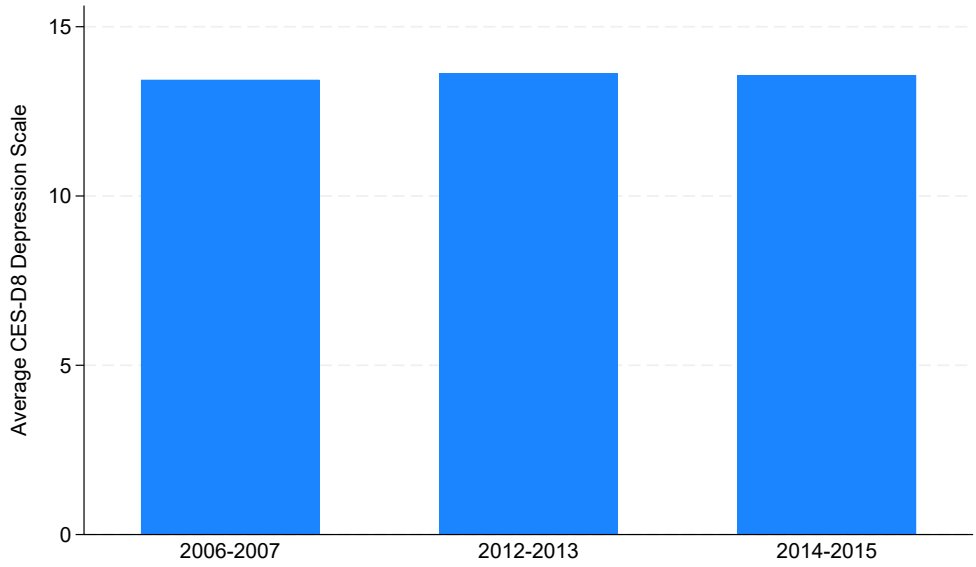

Figure A.1: Average CES-D8 depression scale across waves (second-generation immigrants)

Notes: the figure shows the average CES-D8 depression scale for second-generation immigrants across the ESS waves.

Table A.1: Descriptive statistics

	Full sample			Native			Second-generation		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	N	Mean	SD	N	Mean	SD	N	Mean	SD
CES-D8 depression scale	131,105	13.81	4.203	107,105	13.76	4.192	5,281	13.58	3.968
Felt depressed, how often past week	136,743	1.502	0.702	111,736	1.497	0.699	5,281	1.456	0.666
Felt sad, how often past week	136,737	1.601	0.706	111,741	1.591	0.702	5,281	1.550	0.669
Felt everything did as effort, how often past week	136,733	1.730	0.810	111,767	1.725	0.809	5,281	1.691	0.779
Sleep was restless, how often past week	137,094	1.790	0.852	112,031	1.785	0.846	5,281	1.796	0.865
Felt lonely, how often past week	136,786	1.452	0.734	111,809	1.444	0.729	5,281	1.360	0.658
Could not get going, how often past week	134,577	1.606	0.737	109,834	1.612	0.737	5,281	1.507	0.700
Were happy, how often past week	135,989	2.107	0.822	111,145	2.099	0.817	5,281	2.130	0.821
Enjoyed life, how often past week	136,133	2.126	0.859	111,271	2.120	0.855	5,281	2.090	0.850
Age	137,324	48.34	18.63	112,200	48.72	18.75	5,281	47.20	16.43
Age (squared)	137,324	2,684	1,868	112,200	2,725	1,884	5,281	2,498	1,620
Male	137,722	0.460	0.498	112,493	0.462	0.499	5,281	0.462	0.499
ISCO 08 (1-digit)	123,389	4.753	2.524	100,648	4.767	2.520	5,281	4.261	2.429
Physically hampered	137,858	0.118	0.322	112,605	0.116	0.320	5,281	0.155	0.361
Citizen	137,768	0.957	0.204	112,565	0.997	0.0508	5,281	0.928	0.259
Religiosity	136,574	4.636	3.041	111,594	4.586	3.008	5,281	4.492	3.149
Member of a group discriminated against in this country	136,296	0.0734	0.261	111,520	0.0586	0.235	5,281	0.122	0.327
Paid work	136,906	0.493	0.500	111,839	0.489	0.500	5,281	0.595	0.491
Education	136,906	0.0845	0.278	111,839	0.0832	0.276	5,281	0.0555	0.229
Unemployed	136,906	0.0597	0.237	111,839	0.0570	0.232	5,281	0.0549	0.228
Permanently sick or disabled	136,906	0.0249	0.156	111,839	0.0238	0.153	5,281	0.0318	0.176
Retired	136,906	0.241	0.428	111,839	0.251	0.433	5,281	0.180	0.385
Housework	136,906	0.0848	0.279	111,839	0.0850	0.279	5,281	0.0703	0.256
Other	136,906	0.0118	0.108	111,839	0.0109	0.104	5,281	0.0121	0.109
A big city	137,474	0.217	0.412	112,308	0.193	0.395	5,281	0.309	0.462
Suburbs or outskirts of big city	137,474	0.113	0.316	112,308	0.108	0.310	5,281	0.131	0.337
Town or small city	137,474	0.315	0.464	112,308	0.315	0.465	5,281	0.295	0.456
Country village	137,474	0.298	0.457	112,308	0.320	0.467	5,281	0.227	0.419
Farm or home in countryside	137,474	0.0576	0.233	112,308	0.0634	0.244	5,281	0.0375	0.190
Married	120,817	0.586	0.493	98,252	0.588	0.492	5,281	0.564	0.496
Separated	120,817	0.00114	0.0338	98,252	0.00105	0.0324	5,281	0.00151	0.0389
Widowed	120,817	0.187	0.390	98,252	0.185	0.389	5,281	0.183	0.387
None of these	120,817	0.226	0.418	98,252	0.225	0.418	5,281	0.251	0.434
Less than lower secondary education (ISCED 0-1)	120,050	0.102	0.302	97,533	0.106	0.308	5,281	0.0470	0.212
Lower secondary education completed (ISCED 2)	120,050	0.180	0.384	97,533	0.185	0.389	5,281	0.120	0.325
Upper secondary education completed (ISCED 3)	120,050	0.374	0.484	97,533	0.378	0.485	5,281	0.395	0.489
Post-secondary non-tertiary education completed (ISCED 4)	120,050	0.129	0.336	97,533	0.124	0.329	5,281	0.165	0.371
Tertiary education completed (ISCED 5-6)	120,050	0.215	0.411	97,533	0.207	0.405	5,281	0.273	0.446

Notes: the table presents the descriptive statistics of the variables used in the analysis. The descriptive statistics are: total number of observations (N); mean (Mean); standard deviation (SD) and are divided into Full sample, Natives, and Second-generation immigrants.

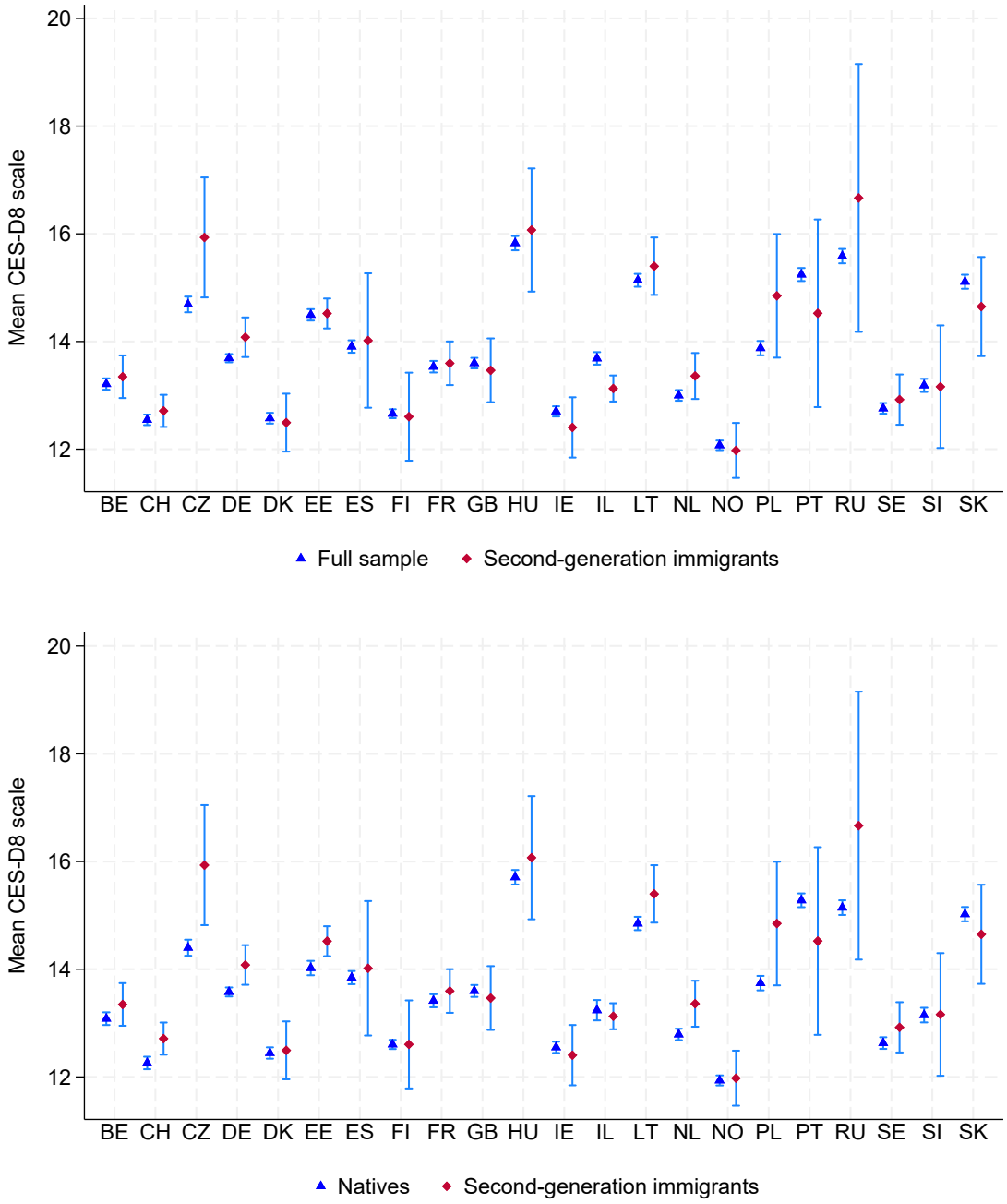

Figure A.2: Average CES-D8 depression scale by country with 95% confidence intervals

Notes: the graph shows differences in the average CES-D8 depression scale by country between the sample of second-generation immigrant and the full sample (above) and between the sample of second-generation immigrant and natives (below). For all three samples, only respondents with no missing values in the eight questions included in the CES-D8 depression scale were included. Confidence intervals at the 95% level are reported.

Table A.2: Descriptive statistics: aggregate variables (country of origin)

Variables	(1) N	(2) Mean	(3) S.D.	(4) Max	(5) Min
Social trust (Standardised) (CO)	5,281	0	1.000	3.181	-2.159
Democracy (CO)	5,281	0.507	0.500	1	0
Trust in parliament (CO)	5,281	2.770	0.199	3.349	1.402
Latent GDP per capita (CO)	5,281	12.16	6.067	27.69	1.179
Health inequality (CO)	5,281	1.576	1.339	3.316	-1.812
Religiosity (CO)	5,281	0.587	0.234	0.995	0.137
Infant mortality (per 1000 live births) (CO)	5,281	63.11	57.59	218.8	10.75
Individualism (CO)	5,281	56.34	18.66	91	8
Subjective health (CO)	5,281	2.306	0.264	2.729	1.729

Notes: the table presents the descriptive statistics of the variables used in the analysis. The descriptive statistics are: total number of observations (1); mean (2); standard deviation (3); maximum value (4); minimum value (5). "CO" stands for "Country of origin".

Table A.3: List countries of origin (ISO2)

ISO2	Freq.	Percen	Cum	**ISO2**	Freq.	Percen	Cum
AR	29	0.55	0.55	KR	4	0.08	50.56
AT	131	2.48	3.03	LT	30	0.57	51.13
AU	8	0.15	3.18	LU	7	0.13	51.26
BD	3	0.06	3.24	LV	45	0.85	52.11
BE	66	1.25	4.49	MA	397	7.52	59.63
BG	34	0.64	5.13	MT	1	0.02	59.65
BR	23	0.44	5.57	MX	4	0.08	59.72
CA	27	0.51	6.08	MY	3	0.06	59.78
CH	25	0.47	6.55	NL	85	1.61	61.39
CL	12	0.23	6.78	NO	55	1.04	62.43
CN	11	0.21	6.99	NZ	2	0.04	62.47
CO	4	0.08	7.06	PE	4	0.08	62.54
CZ	103	1.95	9.01	PH	12	0.23	62.77
DE	498	9.43	18.44	PK	23	0.44	63.21
DK	61	1.16	19.6	PL	395	7.48	70.69
EC	1	0.02	19.62	PT	66	1.25	71.94
EE	11	0.21	19.83	RO	192	3.64	75.57
ES	115	2.18	22	RU	910	17.23	92.8
FI	112	2.12	24.12	SE	66	1.25	94.05
FR	241	4.56	28.69	SG	2	0.04	94.09
GB	190	3.6	32.29	SI	6	0.11	94.21
GR	40	0.76	33.04	TH	6	0.11	94.32
HU	115	2.18	35.22	TR	147	2.78	97.1
ID	86	1.63	36.85	TT	1	0.02	97.12
IE	98	1.86	38.7	TW	1	0.02	97.14
IN	73	1.38	40.09	US	127	2.4	99.55
IR	83	1.57	41.66	UY	5	0.09	99.64
IT	457	8.65	50.31	VE	6	0.11	99.75
JP	9	0.17	50.48	VN	13	0.25	100

Notes: the table presents the full list of countries of origin in the final sample. Each country of origin is labelled with the the ISO2 code.

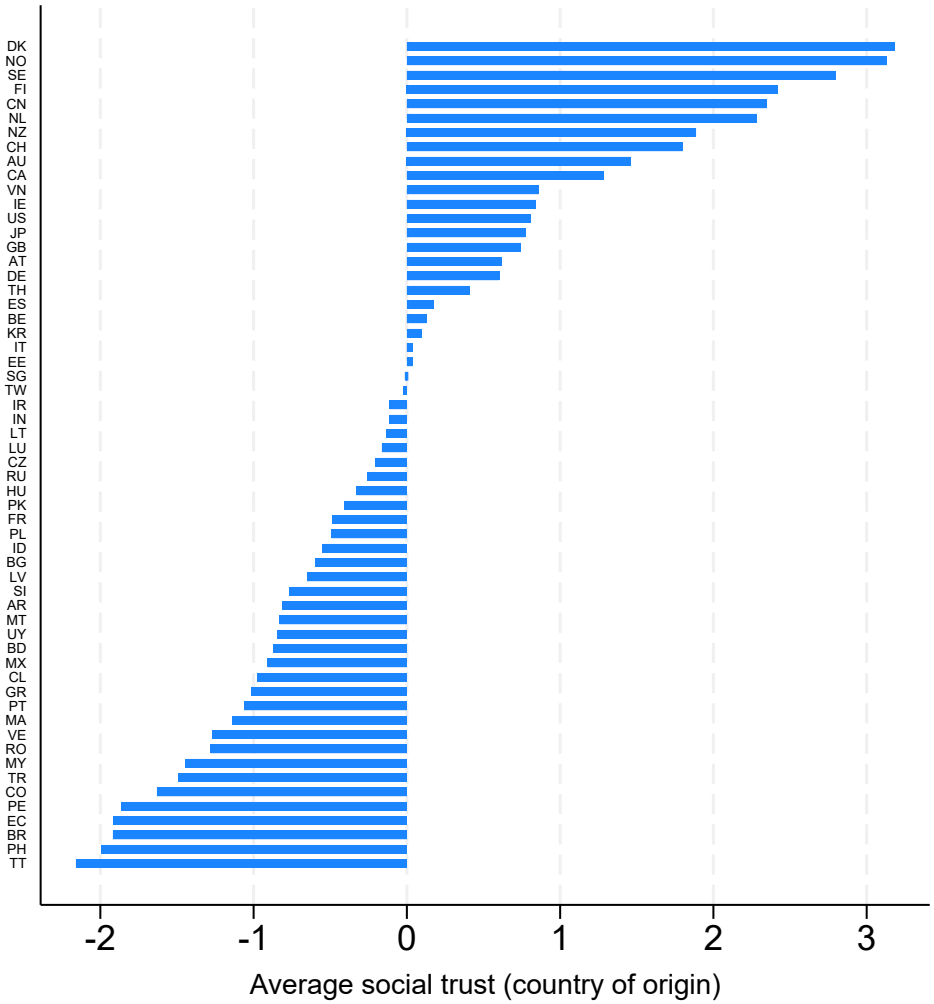

Figure A.3: Average social trust by country of origin

Notes: the figure shows the standardised average inherited social trust by country of origin.

B Mother as main reference

Table B.1: List countries of origin (mother as main reference)

ISO2	Freq.	Percen	Cum	**ISO2**	Freq.	Percen	Cum
AR	33	0.62	0.62	KR	4	0.08	50.61
AT	143	2.7	3.32	LT	27	0.51	51.12
AU	7	0.13	3.45	LU	8	0.15	51.27
BD	3	0.06	3.51	LV	42	0.79	52.06
BE	65	1.23	4.73	MA	384	7.24	59.3
BG	30	0.57	5.3	MT	1	0.02	59.32
BR	24	0.45	5.75	MX	3	0.06	59.38
CA	26	0.49	6.24	MY	2	0.04	59.42
CH	23	0.43	6.67	NL	82	1.55	60.96
CL	10	0.19	6.86	NO	57	1.07	62.04
CN	12	0.23	7.09	NZ	2	0.04	62.07
CO	4	0.08	7.16	PE	5	0.09	62.17
CZ	101	1.9	9.07	PH	14	0.26	62.43
DE	508	9.58	18.64	PK	24	0.45	62.88
DK	60	1.13	19.77	PL	385	7.26	70.14
EC	1	0.02	19.79	PT	67	1.26	71.4
EE	11	0.21	20	RO	188	3.54	74.95
ES	117	2.21	22.21	RU	939	17.7	92.65
FI	110	2.07	24.28	SE	69	1.3	93.95
FR	242	4.56	28.84	SG	2	0.04	93.99
GB	195	3.68	32.52	SI	6	0.11	94.1
GR	42	0.79	33.31	TH	8	0.15	94.25
HU	114	2.15	35.46	TR	151	2.85	97.1
ID	84	1.58	37.04	TT	1	0.02	97.12
IE	102	1.92	38.96	TW	1	0.02	97.13
IN	64	1.21	40.17	US	122	2.3	99.43
IR	87	1.64	41.81	UY	6	0.11	99.55
IT	454	8.56	50.37	VE	7	0.13	99.68
JP	9	0.17	50.54	VN	17	0.32	100

Notes: the table presents the full list of countries of origin with mother as main reference. Each country of origin is labelled with the the ISO2 code.

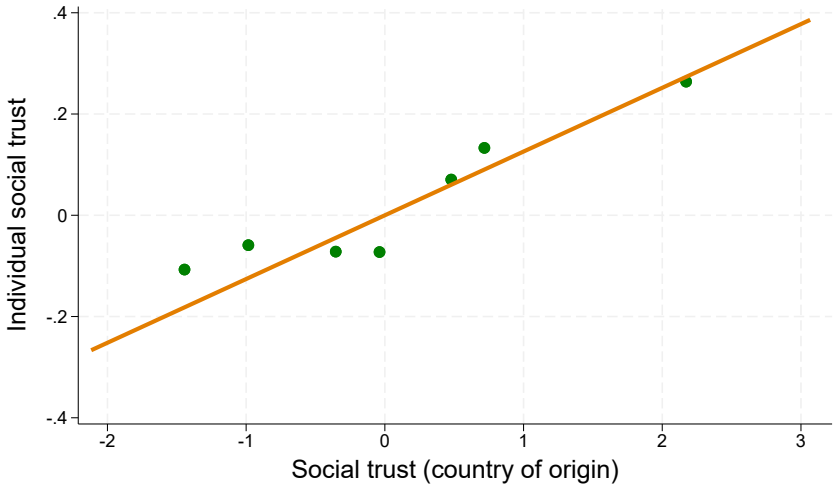

Figure B.1: Binned scatterplot of individual social trust of second-generation immigrants regressed on social trust in the parents' country of origin (mother as main reference).

Notes: the figure shows the binned scatterplot of individual social trust of second-generation immigrants regressed on social trust in the parents' country of origin against age, age (squared), male dummy and residence country-by-wave fixed effects. Parents' country of origin is defined using mother as the main reference. Individual social trust is measured on a scale going from 0 (You can't be too careful) to 10 (Most people can be trusted). Both social trust variables have been standardised with a mean of 0 and a standard deviation of 1. The coefficient of the regression is 0.063 (p-value<0.01). The adjusted-R^2 is 0.0880. The number of bins in the scatterplot is computed using the integrated mean square error-optimal number of bins as suggested in Cattaneo et al. (2024). Binned scatterplot generated using the Stata command `binsreg` (Cattaneo et al., 2019)

Table B.2: Regression: CES-D8 depression scale and social trust in the country of origin (CO) with mother as main reference

	CES-D8 depression scale			
Variables	(1)	(2)	(3)	(4)
Social trust (CO)	-0.058***	-0.042***	-0.085***	-0.075***
	(0.018)	(0.015)	(0.029)	(0.026)
Country-wave FE	✓	✓	✓	✓
Continent FE	✓	✓	✓	✓
Individual Controls		✓		✓
Country-of-origin controls			✓	✓
Country of origin	58	58	58	58
Observations	5,305	5,305	5,305	5,305
Adjusted R-squared	0.059	0.155	0.059	0.155

Notes: Columns (1)–(4) report the depression symptoms regressed on social trust in the country of origin (CO), age, age squared and a male dummy (1), other individual controls (occupation, employment, the highest level of completed education (ISCED), type of respondent's domicile, marital status, degree of religiosity, a citizen dummy, a discrimination dummy, and a hampered-by-health-problems dummy) (2) and country-of-origin controls (latent GDP per capita, health inequality index, infant mortality rate, democracy dummy, individualism, religiosity, trust in the parliament, self-reported health) and both individual and aggregate controls (4). Country-by-wave fixed effects are always included. The abbreviation "CO" stands for "Country of origin". Mother is the main reference to define parents' country of origin. Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

Table B.3: Regression: severity of depressive symptoms and social trust in the country of origin (CO) with mother as main reference category

	CES-D8	**Depressive symptoms**							
	Depression scale	Felt depressed	Felt everything did as effort	Sleep was restless	Felt lonely	Felt sad	Could not get going	Were happy (reversed)	Enjoyed life (reversed)
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Social trust (CO)	-0.058***	-0.074***	-0.017	-0.016	-0.031*	-0.068***	-0.041**	-0.045**	-0.024*
	(0.018)	(0.013)	(0.016)	(0.020)	(0.018)	(0.015)	(0.016)	(0.017)	(0.013)
Country-wave FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country of origin	58	58	58	58	58	58	58	58	58
Observations	5,305	5,305	5,305	5,305	5,305	5,305	5,305	5,305	5,305
Adjusted R-squared	0.059	0.053	0.035	0.025	0.038	0.066	0.059	0.039	0.045

Notes: Columns (1)–(9) report measures for the severity of depression regressed against social trust in the country of origin. The dependent variable is the CES-D8 depression scale in Column (1). Columns (2)-(9) present the effect of social trust in the country of origin on the different symptoms included in the CES-D8 depression scale. The name of the symptom is written on top of each column. In each regression, I control for age, age squared and a male dummy. The abbreviation "CO" stands for "Country of origin". Country-by-wave fixed effects are always included. Mother is the main reference to define parents' country of origin. Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

C Other model specifications

Table C.1: Regression: CES-D8 depression scale and social trust in the country of origin (CO) controlling for income

	CES-D8 depression scale			
Variables	(1)	(2)	(3)	(4)
Social trust (CO)	-0.074***	-0.058***	-0.118***	-0.091***
	(0.021)	(0.019)	(0.034)	(0.030)
Country-wave FE	✓	✓	✓	✓
Individual Controls		✓		✓
Country-of-origin controls			✓	✓
Country of origin	57	57	57	57
Observations	3,481	3,481	3,481	3,481
Adjusted R-squared	0.054	0.163	0.054	0.163

Notes: Columns (1)–(4) report the depression symptoms regressed on social trust in the country of origin (CO), age, age squared and a male dummy (1), other individual controls (income, occupation, employment, the highest level of completed education (ISCED), type of respondent's domicile, marital status, degree of religiosity, a citizen dummy, a discrimination dummy, and a hampered-by-health-problems dummy) (2) and country-of-origin controls (latent GDP per capita, health inequality index, infant mortality rate, democracy dummy, individualism, religiosity, trust in the parliament, self-reported health) and both individual and aggregate controls (4). Only waves 6 and 7 are included. Russia and Slovakia were excluded because they did not have observations in both waves 6 and 7. Country-by-wave fixed effects are always included. The abbreviation "CO" stands for "Country of origin". Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

Table C.2: Depressive symptoms and social trust (CO) with controls

	Felt depressed			
	(1)	(2)	(3)	(4)
Social trust (CO)	-0.080***	-0.070***	-0.103***	-0.097***
	(0.014)	(0.013)	(0.024)	(0.024)
Adjusted R-squared	0.053	0.108	0.053	0.107
	Felt lonely			
	(1)	(2)	(3)	(4)
Social trust (CO)	-0.036**	-0.031**	-0.068***	-0.062***
	(0.018)	(0.015)	(0.023)	(0.020)
Adjusted R-squared	0.039	0.106	0.039	0.106
	Felt sad			
	(1)	(2)	(3)	(4)
Social trust (CO)	-0.068***	-0.055***	-0.080***	-0.074***
	(0.018)	(0.016)	(0.024)	(0.022)
Adjusted R-squared	0.064	0.106	0.064	0.105
	Could not get going			
	(1)	(2)	(3)	(4)
Social trust (CO)	-0.045***	-0.035***	-0.072***	-0.060**
	(0.014)	(0.013)	(0.025)	(0.023)
Adjusted R-squared	0.060	0.105	0.060	0.105
	Were unhappy			
	(1)	(2)	(3)	(4)
Social trust (CO)	-0.055***	-0.050***	-0.063*	-0.057*
	(0.018)	(0.017)	(0.032)	(0.029)
Adjusted R-squared	0.040	0.080	0.039	0.079
	Did not enjoy life			
	(1)	(2)	(3)	(4)
Social trust (CO)	-0.032***	-0.029**	-0.046	-0.039
	(0.012)	(0.011)	(0.036)	(0.034)
Adjusted R-squared	0.047	0.082	0.048	0.082
Country of origin	58	58	58	58
Observations	5,281	5,281	5,281	5,281

Notes: Columns (1)-—(4) report the depression symptoms regressed on social trust in the country of origin (CO), age, age squared and male (1), other individual controls (2) country-of-origin controls (3) and other individual controls and country-of-origin controls (4). Age, age (squared), male dummy and country-by-wave fixed effects are always included. The abbreviation "CO" stands for "Country of origin". Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

Table C.3: Regression: severity of depressive symptoms and social trust in the country of origin (CO) with aggregate variables

	CES-D8 depression scale							
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Social trust (CO)	-0.057**	-0.069***	-0.055**	-0.064***	-0.060***	-0.054**	-0.071***	-0.082***
	(0.024)	(0.021)	(0.022)	(0.021)	(0.021)	(0.022)	(0.018)	(0.020)
GDP per capita (CO)	-0.002							
	(0.003)							
Democracy (CO)		0.020						
		(0.033)						
Religiosity (CO)			0.071					
			(0.062)					
Individualism (CO)				-0.000				
				(0.001)				
Health inequality (CO)					-0.008			
					(0.010)			
Infant mortality (CO)						0.000		
						(0.000)		
Subjective health (CO)							-0.085	
							(0.065)	
Trust in parliament (CO)								-0.196**
								(0.086)
Country-wave FE	✓	✓	✓	✓	✓	✓	✓	✓
Country of origin	58	58	58	58	58	58	58	58
Observations	5,281	5,281	5,281	5,281	5,281	5,281	5,281	5,281
Adjusted R-squared	0.058	0.058	0.058	0.058	0.058	0.059	0.059	0.059

Notes: Columns (1)–(8) report the depression symptoms (CES-D8 depression scale) regressed on social trust in the country of origin (CO), age, age squared and gender and GDP per capita (CO) (1), democracy (CO) (2), religiosity (CO), individualism (CO), health inequality (CO), infant mortality (CO), subjective health (CO), trust in parliament (CO). Country-by-wave fixed effects are always included. The abbreviation "CO" stands for "Country of origin". Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

Table C.4: Regression: severity of depressive symptoms on social trust in the country of origin (CO) using ordered probit and ordered logit

	Ordered probit							
	Depressive symptoms							
Panel (a)	Felt depressed (1)	Felt everything did as effort (2)	Sleep was restless (3)	Felt lonely (4)	Felt sad (5)	Could not get going (6)	Were happy (reversed) (7)	Enjoyed life (reversed) (8)
Social trust (CO)	-0.110***	-0.026	-0.021	-0.062**	-0.095***	-0.057***	-0.062***	-0.034***
	(0.021)	(0.020)	(0.025)	(0.028)	(0.024)	(0.018)	(0.021)	(0.013)
Country-wave FE	✓	✓	✓	✓	✓	✓	✓	✓
Country of origin	58	58	58	58	58	58	58	58
Observations	5,281	5,281	5,281	5,281	5,281	5,281	5,281	5,281
	Ordered logit							
	Depressive symptoms							
Panel (b)	Felt depressed (1)	Felt everything did as effort (2)	Sleep was restless (3)	Felt lonely (4)	Felt sad (5)	Could not get going (6)	Were happy (reversed) (7)	Enjoyed life (reversed) (8)
Social trust (CO)	-0.188***	-0.045	-0.041	-0.116**	-0.160***	-0.091***	-0.099***	-0.053**
	(0.036)	(0.037)	(0.043)	(0.053)	(0.041)	(0.031)	(0.035)	(0.023)
Country-wave FE	✓	✓	✓	✓	✓	✓	✓	✓
Country of origin	58	58	58	58	58	58	58	58
Observations	5,281	5,281	5,281	5,281	5,281	5,281	5,281	5,281

Notes: Columns (1)–(8) report measures for the severity of depressive symptoms regressed against social trust in the country of origin using ordered probit (Panel (a)) and ordered logit (Panel (b)). The name of the symptom is written on top of each column. In each regression, I control for age, age squared and a male dummy. The abbreviation "CO" stands for "Country of origin". Country-by-wave fixed effects are always included. Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

D Robustness checks

D.1 Cluster heterogeneity

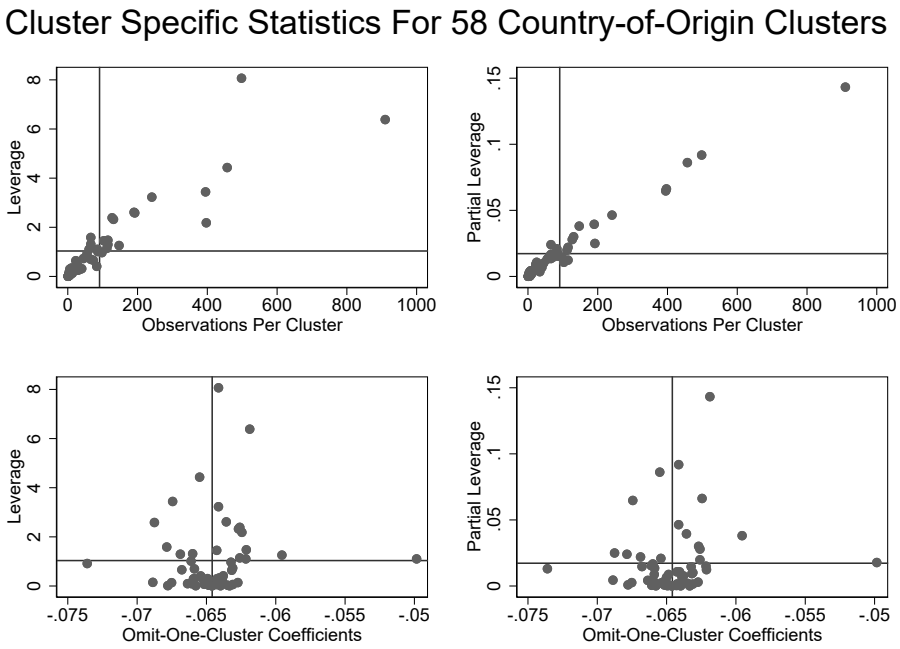

Figure D.1: Leverage and partial leverage clusters

Notes: the figure shows the plots for leverage and partial leverage clusters of the CES-D8 depression scale regressed against social trust in the country of origin.

D.2 Social trust volatility

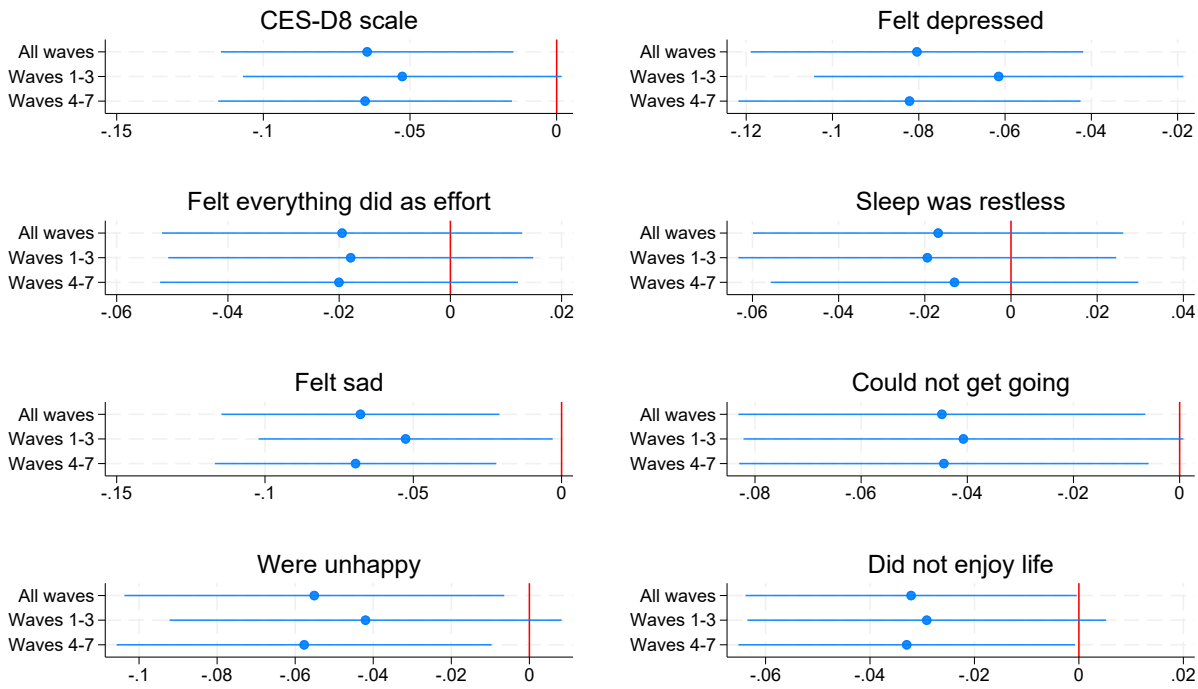

Figure D.2: Estimation with different waves

Notes: the graph shows the coefficient for social trust in Table 1 compared with the coefficients in Table D.1. Confidence intervals are at the 1% level if coefficient in Table 1 is statistically significant at the 1% level; otherwise confidence intervals are at the 5% level.

Table D.1: Regression: severity of depressive symptoms on social trust in the country of origin (CO) by waves groups

	Waves 1 – 3								
	CES-D8	**Depressive symptoms**							
	Depression scale	Felt depressed	Felt everything did as effort	Sleep was restless	Felt lonely	Felt sad	Could not get going	Were happy (reversed)	Enjoyed life (reversed)
Panel (a)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Social trust (CO)	-0.053**	-0.061***	-0.018	-0.019	-0.022	-0.053***	-0.041**	-0.042**	-0.029**
	(0.020)	(0.016)	(0.016)	(0.022)	(0.017)	(0.018)	(0.015)	(0.019)	(0.013)
Country-wave FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country of origin	47	47	47	47	47	47	47	47	47
Observations	4,737	4,737	4,737	4,737	4,737	4,737	4,737	4,737	4,737
Adjusted R-squared	0.062	0.053	0.038	0.026	0.038	0.069	0.062	0.040	0.051

	Waves 4 – 7								
	CES-D8	**Depressive symptoms**							
	Depression scale	Felt depressed	Felt everything did as effort	Sleep was restless	Felt lonely	Felt sad	Could not get going	Were happy (reversed)	Enjoyed life (reversed)
Panel (b)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Social trust (CO)	-0.065***	-0.082***	-0.020	-0.013	-0.037**	-0.069***	-0.044***	-0.058***	-0.033***
	(0.019)	(0.015)	(0.016)	(0.021)	(0.017)	(0.018)	(0.014)	(0.018)	(0.012)
Country-wave FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country of origin	58	58	58	58	58	58	58	58	58
Observations	5,281	5,281	5,281	5,281	5,281	5,281	5,281	5,281	5,281
Adjusted R-squared	0.059	0.054	0.034	0.025	0.039	0.064	0.060	0.040	0.047

Notes: Columns (1)–(9) report the depression symptoms regressed on social trust in the country of origin (CO). Panel (a) includes social trust in the country of origin averaged only across waves 1-3 of the IVS. Panel (b) includes social trust in the country of origin averaged only across waves 4-7 of the IVS. Age, age squared, male dummy and country-by-wave fixed effects are always included. The abbreviation "CO" stands for "Country of origin". Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

Table D.2: Regression: severity of depressive symptoms and social trust in the country of origin (CO) including only CO in waves 1-3

	CES-D8	**Depressive symptoms**							
	Depression scale	Felt depressed	Felt everything did as effort	Sleep was restless	Felt lonely	Felt sad	Could not get going	Were happy (reversed)	Enjoyed life (reversed)
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Social trust (CO)	-0.055**	-0.076***	-0.009	-0.018	-0.028	-0.060***	-0.036**	-0.051**	-0.023
	(0.022)	(0.018)	(0.020)	(0.024)	(0.018)	(0.020)	(0.014)	(0.022)	(0.015)
Country-wave FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country of origin	47	47	47	47	47	47	47	47	47
Observations	4,737	4,737	4,737	4,737	4,737	4,737	4,737	4,737	4,737
Adjusted R-squared	0.065	0.058	0.039	0.031	0.041	0.072	0.056	0.042	0.053

Notes: Columns (1)–(9) report measures for the severity of depression regressed against social trust in the country of origin. The dependent variable is the CES-D8 depression scale in Column (1). Columns (2)-(9) present the effect of social trust in the country of origin on the different symptoms included in the CES-D8 depression scale. The name of the symptom is written on top of each column. In each regression, I control for age, age squared and a male dummy. The abbreviation "CO" stands for "Country of origin". Country-by-wave fixed effects are always included. The sample size is the same as in table **??**. Standard errors clustered at the country of origin level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.